

Design and Implementation of a Flight Controller for a Quadrotor Unmanned Aerial Vehicle

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Abstract:

This work aims to develop a low-cost, open-source quadrotor flight controller using ESP32 to address the limitations of proprietary systems and complex open-source platforms. A custom PCB integrates IMU, GPS, barometer, and Wi-Fi, with dual-loop PID control and a complementary filter for stable flight and accurate estimation. Experimental results demonstrate stable hover and responsive control ($\zeta = 0.71$, $PM = 81^\circ$). The system provides a cost-effective, scalable solution for UAV education, research, and future applications such as autonomous navigation and intelligent control.

Introduction:

Background and Context:

UAVs are widely used in many fields, increasing the need for efficient flight controllers. However, existing solutions are either costly and proprietary or too complex for educational use.

Problem Statement:

There is a lack of a low-cost, open-source, and easy-to-understand flight controller suitable for students and practical learning.

Objectives and Scope:

This work aims to develop an ESP32-based quadrotor flight controller with integrated sensors, PID control, and wireless telemetry. It focuses on a cost-effective, simple, and scalable platform for UAV education and experimentation.

Objectives:

- Design an ESP32-based quadrotor flight controller.
- Develop sensor integration (IMU, GPS, barometer).
- Implement PID control for stabilization.
- Apply sensor fusion using a complementary filter.
- Evaluate system performance through testing.

Proposed Work:

Figure 1 illustrates the proposed ESP32-based quadrotor flight controller implemented on a custom PCB integrating IMU (MPU6050), GPS module, barometer, and nRF24L01 telemetry. The receiver provides pilot inputs to the ESP32, which processes sensor data and generates PWM signals to control ESCs and motors. A dual-loop PID control system ensures stable flight, while a complementary filter improves attitude estimation. The ESP32's dual-core architecture enables reliable real-time control and communication. In this paper, the black box is a custom logging system that leverages the ESP32's dual-core architecture to record time-synchronized flight data (such as sensor measurements and PID outputs) into a circular RAM buffer for post-flight analysis and tuning without disrupting the real-time control loop.

Novelty: Low-cost, open-source design with integrated Wi-Fi, simplified architecture, and efficient sensor-actuator integration suitable for education and research.

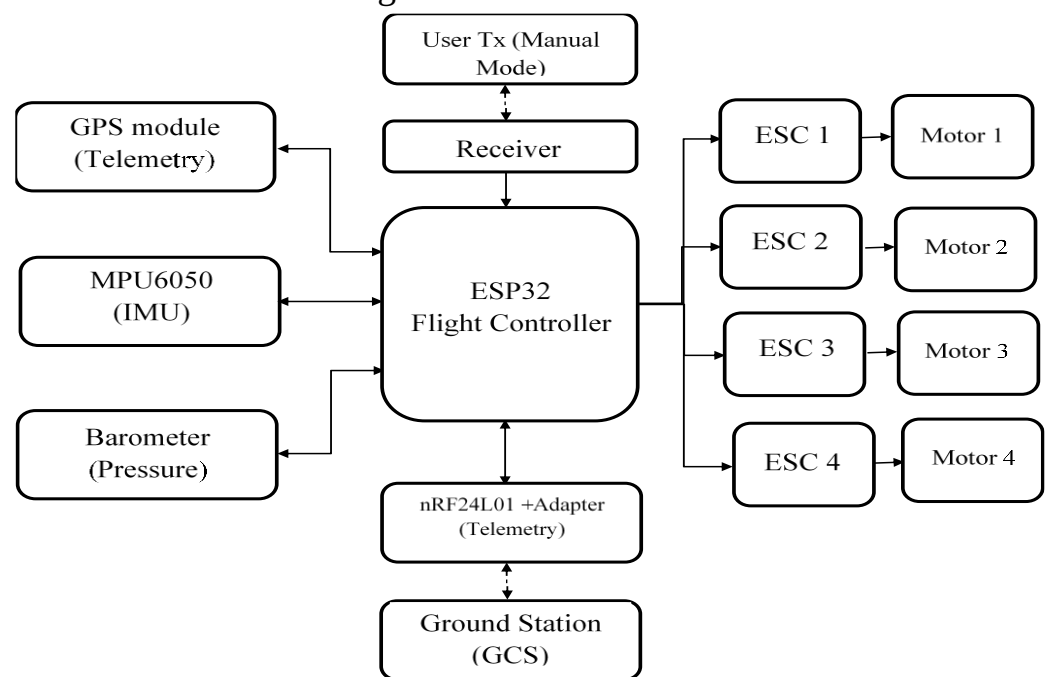


Figure 1 – Block diagram of drone flight controller system

Feature:

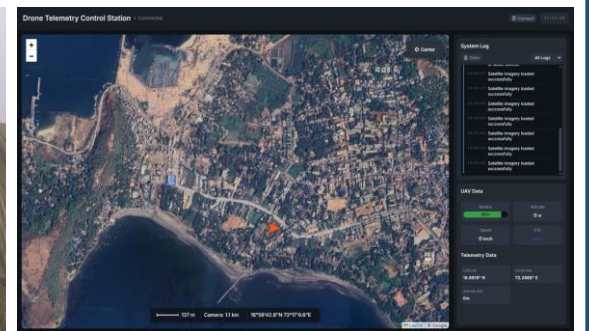
- Dual-core ESP32 processing
- Integrated Wi-Fi and telemetry
- Multi-sensor support (IMU, GPS, barometer)
- Real-time data monitoring
- Custom PCB design

Applications:

- Educational UAV platform
- Drone research and development
- Precision agriculture
- Surveillance and monitoring
- Environmental studies
- Autonomous UAV systems

Results:

The quadrotor achieved stable hover and responsive control, with JSON-based real-time monitoring of GPS, battery, altitude, speed, and position



at the Ground Control Station (Fig. 2 & Fig. 3).

Conclusion:

The system achieves stable and reliable flight with accurate control and sensor fusion. Its low-cost, open-source design makes it suitable for educational and research applications, with scope for future enhancements like autonomous navigation and advanced control.